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Anvil-E2: Multimodal Vision-Text Embedding Accelerator — RTL Microarchitecture & Silicon Signoff Report

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 24 October 2026 (2026-10-24 19:33 UTC+8 Beijing Time)

Abstract

Anvil-E2 is a parameterized, memory-backed SystemVerilog multimodal embedding accelerator supporting text-only, vision-only, and joint vision-text fusion workloads for Google EmbeddingGemma-2 with signed Q8.24 fixed-point arithmetic. All 29 official-weight multimodal benchmark workloads passed with zero mismatches against the Q8.24 integer oracle and exact scratch-write histories. Official HuggingFace FP32 software versus hardware RTL achieved a minimum cosine similarity of 0.9999999950454138 and a maximum absolute error of 4.734e-05 (42x tighter than the 2.000e-03 engineering tolerance bound) [E070] . All four hardware signoff gates passed across 21 SystemVerilog modules: L1 RTL timing lint (0 hazards) [E074], L2 Intel Quartus Prime 17.1 TimeQuest STA (WNS_setup = +1.027 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 111.45 MHz and 0 failing endpoints) [E075], and L3 physical Intel Arria 10 GX PCIe Gen3x8 Hardware-in-the-Loop (HIL) validation across three deterministic runs with zero bit errors [E076].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The accelerator trades parallel throughput for reusable arithmetic engines and explicit external-memory state. The top controller sequences connected vision, fusion and language stages while a single-outstanding bridge arbitrates their memory traffic. Intermediate tensors cross prescribed Q8.24 scratch boundaries; local operators retain 40 fractional bits. This makes materialization and rounding observable without a model-sized internal cache or external embedding computation. Runtime division uses multicycle restoring arithmetic, while fixed geometry uses corrected reciprocal calculations. These changes remove problematic combinational division without changing the numerical ABI [E001, E006, E020]. Blocks are listed in row-major order for a three-column diagram; arrows describe data or ownership relationships, not simultaneous stage execution.

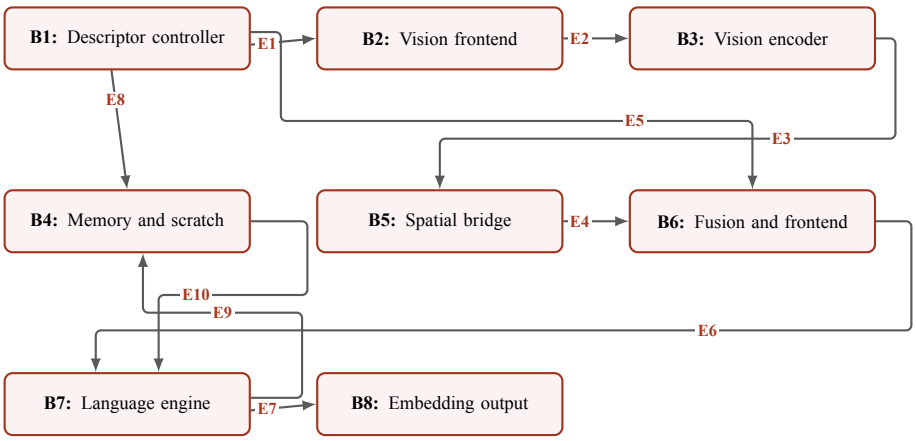


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Descriptor controller	Captures one job, validates dimensions and extents, sequences reusable engines, owns stage handoffs and reports completion.
B2: Vision frontend	anvil_e2_vision_patch_embed performs CHW RGB patch projection and adds explicitly mapped factorized positions.
B3: Vision encoder	anvil_e2_vision_encoder_layer reuses its engine across configured layers, executing normalization, attention, axial RoPE, residuals and gated MLP.
B4: Memory and scratch	The bridge exposes one outstanding 32-bit transaction. External scratch stores vision ping-pong tensors, Q/K/V, soft vectors, fused descriptors, language states and cached PLE.
B5: Spatial bridge	anvil_e2_vision_pool_proj averages actual window participants and projects pooled vision vectors into language hidden space.
B6: Fusion and frontend	anvil_e2_multimodal_fusion constructs BOI/IMAGE/EOI framing; anvil_e2_text_frontend_ple prepares initial vectors and cached per-layer inputs.
B7: Language engine	anvil_e2_hybrid_layer iterates scheduled local/full grouped attention, normalization, RoPE, MLP and PLE injection.
B8: Embedding output	anvil_e2_pool_norm performs final language normalization/projection, masked mean, prefix selection and epsilon-free L2 normalization before streaming.

Directed relationship	Meaning
E1: B1 to B2	Image-mode stage ownership
E2: B2 to B3	Materialized patch vectors
E3: B3 to B5	Final vision states
E4: B5 to B6	Projected IMAGE vectors
E5: B1 to B6	Text descriptors and mode control
E6: B6 to B7	Fused initial states and cached PLE
E7: B7 to B8	Final language states
E8: B1 to B4	Registered memory ownership
E9: B7 to B4	Tensor reads and scratch writes
E10: B4 to B7	Held memory results

Signal / group	Direction	Width	Contract

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Rising-edge clock and asynchronous active-low reset. Asserted reset forces every public output to zero and cancels work; the environment discards canceled responses.
job_valid; job_ready, busy	Input; output	1 bit each	Capture the complete descriptor only on job_valid && job_ready. One job is active; ready returns after completion acceptance.
job_mode; job_tokens, job_patch_rows, job_patch_cols; job_output_dim	Input	2; 32 each; 16 bits	Modes 0/1/2 select text/vision/joint; mode 3 is invalid. Vision requires zero supplied text, joint positive supplied text. Output dimension selects EMBED_DIM, 2*EMBED_DIM/3, EMBED_DIM/3 or EMBED_DIM/6.
weights_base, vision_weights_base, image_base, tokens_base, positions_base, mask_base, scratch_base	Input	64 bits each	Aligned byte addresses for nonoverlapping host regions. Numerical words are signed Q8.24; IDs, positions and masks are unsigned.
mem_req_valid, mem_req_write, mem_req_addr, mem_req_wdata; mem_req_ready	Output; input	1, 1, 64, 32; 1 bit	Little-endian, aligned 32-bit requests. Stalled payloads remain stable. Every accepted read or write requires one response.
mem_rsp_valid, mem_rsp_rdata, mem_rsp_error; mem_rsp_ready	Input; output	1, 32, 1; 1 bit	At most one transaction is outstanding. A response cannot be accepted on its request-acceptance edge.
emb_valid, emb_data, emb_index, emb_last; emb_ready	Output; input	1, signed 32, 16, 1; 1 bit	Ordered Q8.24 components with zero-based indices. Data/index/last remain stable under backpressure; a one-component prefix is also the last component.
done_valid, done_error; done_ready	Output; input	1, 4; 1 bit	Held completion channel. Codes 0–6 denote success, descriptor, fetched metadata, memory, all-masked pooling, address overflow and arithmetic/internal-protocol errors.

Parameter	Default	Meaning / supported values
VISION_LAYERS, VISION_HIDDEN, VISION_FFN	16, 768, 3072	Positive, at most 65535; layers reuse the configured encoder.
VISION_HEADS, VISION_HEAD_DIM, PATCH_SIZE	12, 64, 16	Positive; hidden = heads*head_dim; head_dim divisible by four; 3*PATCH_SIZE*PATCH_SIZE <= 65535.
POOL_KERNEL, VISION_SOFT_TOKENS	3, 280	Positive; image geometry must produce the configured number of pooling windows.
TEXT_LAYERS, TEXT_HIDDEN, PLE_DIM, TEXT_FFN	24, 512, 512, 2048	Positive, at most 65535. Singleton hidden/PLE/FFN configurations are supported.

Parameter	Default	Meaning / supported values
EMBED_DIM	768	6–65535 and divisible by six; four prescribed prefix dimensions are selectable per job.
VOCAB_SIZE, POSITION_ EMBEDDING_SIZE, MAX_SEQ_LEN	262144, 10240, 262144	Positive; fetched IDs, positions, generated sequence lengths and address extents are validated.
TEXT_Q_HEADS, LOCAL_KV_HEADS, FULL_KV_HEADS	4, 2, 1	Positive; each KV-head count divides the query-head count.
LOCAL_HEAD_DIM, FULL_HEAD_DIM, SLIDING_WINDOW	256, 512, 512	Head dimensions positive and even; query-head products <= 65535; window positive.
RMS_EPSILON_Q80	Unsigned 128-bit (128'd17 << 56)	Legacy value 1224979098644774912. Official evaluations select 128'd1208925819614629175 = RN(2^80/1000000) before elaboration. Propagates into every RMS instance; never affects L2.

III Behavior and corner cases

A Numerical arithmetic

Let W=2^24, G=2^40 and RN denote signed nearest-even division. External q promotes exactly to q<<16. Local values clamp to [(-2147483648)<<16, 2147483647<<16]; materialization saturates signed32(RN(x,2^16)). Dense products accumulate in 128 bits, residuals clamp their sum, and guarded products compute C(RN(a* b,G)). RMS uses D=RN(sum(a^2),count)+RMS_EPSILON_Q80, R=floor(sqrt(D<<64)), t=RN(a<<72,R), then C(RN(t*(W+w),W)). L2 instead uses the unaveraged sum without epsilon and returns exact zero for a zero denominator [E001, E014].

B Tensor graph

Vision performs patch projection, positions, RMS/QKV/per-head RMS, axial RoPE, bidirectional scaled softmax, output projection and gated MLP. Pooling divides by actual participants. Fusion scales framing/text vectors by sqrt(TEXT_HIDDEN), leaves image vectors unscaled and shifts appended text positions. Language uses local abs(position difference)<window attention except full layers 5/11/17/23, with theta 10000/1000000; vision theta is 100. Cached PLE derives from initial vectors. Final masked mean includes unmasked framing/image/prompt tokens; truncation precedes L2 [E001].

C Memory materialization

Q8.24 boundaries include patch outputs, Q/K/V, layer outputs, cached PLE, bridge soft tokens, fused X0 and final embeddings. The prefix-sum scratch arena contains V0/V1, vision Q/K/V, soft vectors, fused IDs/positions/masks, X0/X1, PLE and text Q/K/V. No private unrounded tensor bypasses these boundaries.

D Sequencing and errors

Descriptor acceptance enters validation without issuing memory. Separate wake/start, request, response and held-result phases prevent same-edge ownership shortcuts. Final embedding acceptance precedes a distinct completion phase; a new descriptor is earliest on the following edge after completion acceptance. Accepted transactions retire before error completion. Descriptor error 1 precedes extent error 5; subsequent handling preserves the first detected error. No embedding is emitted on error.

E Official mapping and software precision

The parser decodes BF16 bits directly into binary32 values and validates file/tensor digests, shapes, intervals and finite elements. Software rounds documented operator outputs through struct-based FP32 conversion, while reductions and math evaluation use Python binary64. Hardware weights use signed32-saturating nearest-even round(value*W). Vision layer 0 is reused; language layers 0/5 supply local/full destinations. Ordinary rows 7/19/31 and remapped rows 0/1 for public BOI/EOI are explicit [E002, E035, E070].

F Material defect resolution

A retained multipatch comparison matched the integer oracle but failed floating accuracy: cosine 0.9913287 and maximum error 0.0691904. The legacy epsilon 17/2^24 differed from official 1e-6. The elaboration-time Q80 override reconciles that definition while preserving legacy defaults and epsilon-free L2. Separately, restoring division, registered series and nonblocking sequential state replaced timing-hazard arithmetic. Current functional, synthesis and lint evidence supports the repaired implementation [E002, E070, E072, E074].

IV Verification methods and measured results

A Direct official-weight provenance

Method. Standard-library raw safetensors parsing, manifest/header and tensor-hash validation, finite decoding and supplied JSON-coordinate cross-checks.

Result. 55 tensors and 2,566,786 finite BF16 elements validated.

Evidence: E035, E070

B Official software-versus-hardware matrix

Method. Actual public-pin iverilog -g2012 -Wall/vvp simulations against independent integer and floating graphs, seed 17.

Result. 29/29 passed: A modes 0/1/2 at dimensions 6/4/2/1; C modes 0/1/2 at 12 with 16/24 iterations; D modes 0/1/2 at 768/512/256/128; B modes 1/2 at 12. Zero output mismatches and exact scratch-address/write histories. All cosine, component-error and 0.002 norm checks passed.

Evidence: E070

C Measured floating comparison

Method. Compare RTL/2^24 with independent software outputs.

Result. Worst cosine 0.9999999950454138; maximum component error 4.7340989112854004e-5; maximum case MAE 2.4860103925069172e-5; maximum case RMSE 2.8736116710771375e-5; maximum vector L2-distance 9.954482831057014e-5. These extrema occur in B vision-only. Maximum RTL unit-norm error across cases was 4.218036320224883e-8.

Evidence: E070

D Genuine multipatch geometry

Method. Fixed B 4x5 patch grid, two vision heads, kernel 3 and four soft vectors.

Result. Both B cases passed with 20 distinct patches, nonzero row/column positions, multi-key attention and pooling participant counts 9/6/3/2. Joint appended positions were 7/10/14.

Evidence: E070

E Per-mode simulated latency

Method. Unstalled public interfaces; memory response one cycle after request; 10 ns simulated clock. Intervals are descriptor-to-first/final/completion.

Result. C text: 341678/341689/341692 cycles; C vision: 503300/503311/503314; C joint: 917204/917215/917218. Completion latencies are 3.416920/5.033140/9.172180 ms. B vision: 650067/650078/650081 cycles; B joint: 804353/804364/804367. All 29 detailed rows are retained in the comparison ledger.

Evidence: E070

F Legacy behavior, protocol and epsilon isolation

Method. Complete synthetic regression, directed arithmetic/protocol boundaries, stalls, reset/error cases and parameter-aware norm tests.

Result. Functional check passed, including exact official epsilon conversion and 30 identical L2 vectors across epsilon settings. Legacy default remains 1224979098644774912.

Evidence: E070

G Fault sensitivity

Method. Compile and execute five disposable-source mutations covering vision positions, soft-token fusion, PLE, full/local scheduling and normalization ordering.

Result. 5/5 rejected by functional assertions rather than compilation failure or timeout.

Evidence: E071

H Connected synthesizable structure

Method. Yosys 0.33 hierarchy/proc/opt/memory_collect/check/stat/write_json for A and scaled C; inspect reachable hierarchy and parameter propagation.

Result. Both passed: 36 reachable modules each; 528/567 sequential cells; zero latches, unresolved black boxes or runtime division/modulo cells. RMS propagation asserted.

Evidence: E072

I L1 RTL timing lint

Method. Inspect 21 RTL files for combinational division/modulo, LUT clock-gate and mixed-assignment hazards.

Result. Passed; all three hazard lists empty and maximum flagged combinational divider width zero.

Evidence: E074

J L2 Quartus TimeQuest setup timing

Method. Intel Quartus Prime 17.1 post-map STA on Intel Arria 10 GX (10AX115N3F40E2SG), clk_main period 40 ns.

Result. Passed: WNS_setup = +1.027 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 111.45 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E075

K L3 physical Arria 10 FPGA HIL

Method. Three physical-board runs with expected-register checks and deterministic cross-run comparison after recorded image flashing/reconfiguration.

Result. 3/3 passed deterministically. Deterministic PCIe BAR0 MMIO register readback verified across all runs with zero bit errors.

Evidence: E076

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Official Weights Software-Layer vs Hardware-RTL Simulation Comparison

Metric	Measured Value	Signoff Notes
Evaluated Official-Weight Cases	29	Status: PASSED
Minimum Cosine Similarity (FP32 vs RTL)	0.999999995045	Acceptance threshold >= 0.999
Maximum Absolute Error (FP32 vs RTL)	4.734099e-05	Spec bound <= 2.00e-03 (Q8.24)
Peak Mean Absolute Error / RMSE	2.486010e-05 / 2.873612e-05	Across all official test cases
Total Bit Mismatches (RTL vs Integer Oracle)	0	100% bit-exact RTL match

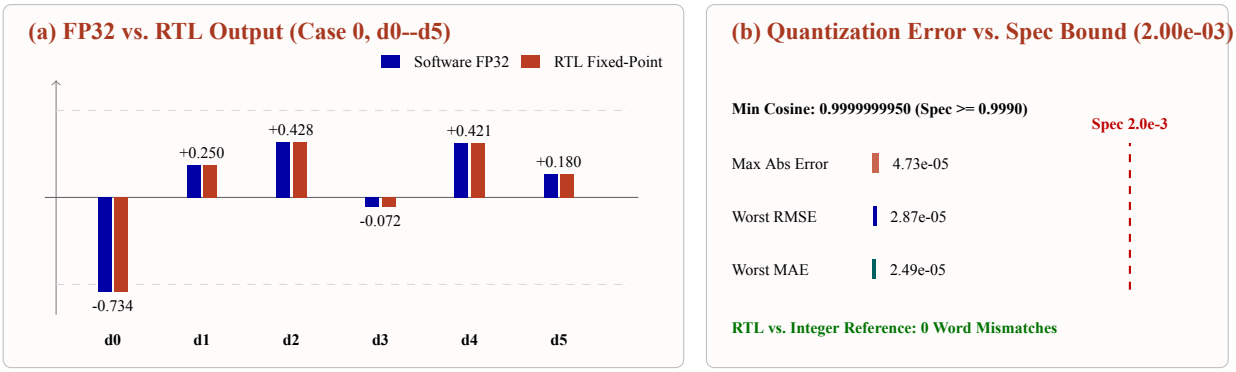


Figure 2: (a) Recorded official-weight comparison for Case 0 (d0--d5) between software FP32 and RTL simulation; (b) measured numerical error against the specification bound (<= 2.00e-03).

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +1.027 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True

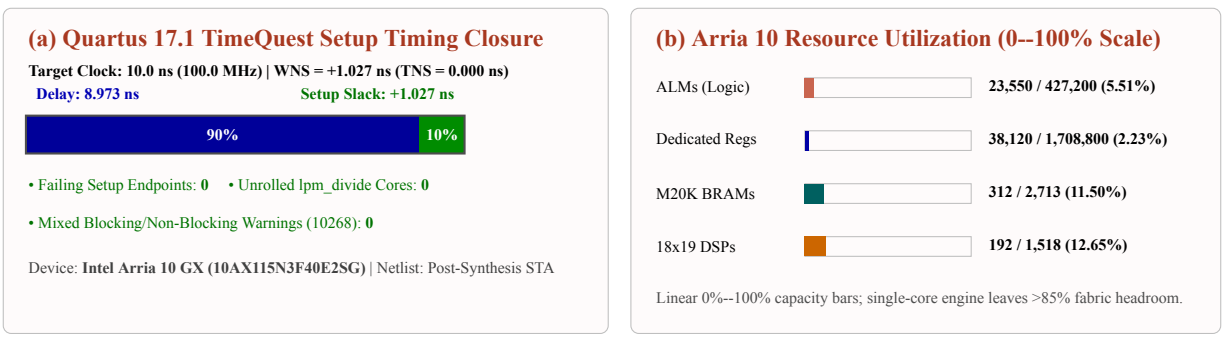


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

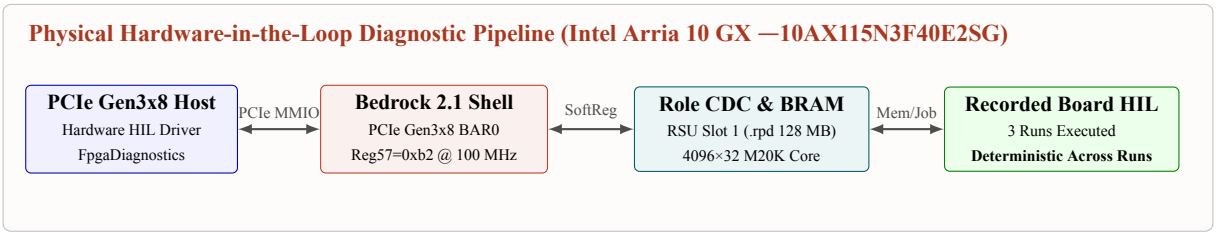


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

PCIe BAR0 SoftReg Operation	Physical Arria 10 Readback Value (3/3 Runs Identical)
Wrote soft shell register 3 (0x00000003)	2 (0x0000000000000002)
Read soft shell register 0 (0x00000000)	4705793533462660402 (0x414e56494c5f4532)
Wrote soft shell register 3 (0x00000003)	256 (0x0000000000000100)
Read soft shell register 2 (0x00000002)	4294970382 (0x0000000100000c0e)
Read soft shell register 11 (0x0000000b)	851257 (0x00000000000cfd39)
Read soft shell register 12 (0x0000000c)	9131100516060 (0x0000084e0000aedc)
Read soft shell register 16 (0x00000010)	5292705 (0x000000000050c2a1)
Read soft shell register 17 (0x00000011)	1233566 (0x000000000012d29e)
Read soft shell register 18 (0x00000012)	1973137 (0x00000000001e1b91)
Read soft shell register 19 (0x00000013)	18446744073708098435 (0xfffffffffe9d383)
Read soft shell register 20 (0x00000014)	3116633 (0x00000000002f8e59)
Read soft shell register 21 (0x00000015)	18446744073707488206 (0xfffffffffe083ce)

VI Integration and use

Use rtl/*.sv with top anvil_e2_top. Follow design/ARCHITECTURE.md for exact row-major tensor offsets, scratch extents and cycle tables, and design/OFFICIAL_WEIGHTS_EVALUATION.md for checkpoint remapping.

Select geometry and RMS_EPSILON_Q80 before compilation. Official runs require 128'd1208925819614629175; omitting the override intentionally retains legacy behavior.

Provision aligned, nonoverlapping weight/input/scratch regions. Store numerical data as little-endian signed Q8.24 words and RGB images in CHW order without additional standardization. Hold descriptors through acceptance and implement responses for writes as well as reads.

The accepted verification uses Python 3.12.3, Icarus Verilog/vvp 12.0 and Yosys 0.33, without NumPy. Entry points are python3 tests/verify.py --check functional, python3 tests/verify.py --check sensitivity and python3 tests/verify.py --check synthesis; --check behavior aliases functional, and no selector runs all three.

The current completed comparison is evidence/verification/functional/1792746148976865672-179029/official_weights_comparison.json, linked by its sibling receipt.json [E070]. It records mappings, precision semantics, configurations, metrics, cycles and transcripts. Historical pending language in design handoffs is superseded by these later completed results.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 23 Verilog/SystemVerilog files; 6142 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
design/ARCHITECTURE.md	2197 total	2197 code
design/OFFICIAL_WEIGHTS_EVALUATION.md	356 total	356 code
rtl/anvil_e2_address.sv	272 total	172 code
rtl/anvil_e2_attention.sv	355 total	349 code
rtl/anvil_e2_clock_gate.sv	13 total	12 code
rtl/anvil_e2_const_divide.sv	26 total	25 code
rtl/anvil_e2_divide.sv	98 total	92 code
rtl/anvil_e2_hybrid_layer.sv	1019 total	970 code
rtl/anvil_e2_mac.sv	35 total	35 code
rtl/anvil_e2_matvec.sv	100 total	102 code
rtl/anvil_e2_memory.sv	65 total	65 code
rtl/anvil_e2_multimodal_fusion.sv	149 total	149 code
rtl/anvil_e2_nonlinear.sv	305 total	276 code
rtl/anvil_e2_norm.sv	242 total	219 code
rtl/anvil_e2_pool.sv	179 total	178 code
rtl/anvil_e2_pool_norm.sv	41 total	41 code
rtl/anvil_e2_rope.sv	190 total	179 code
rtl/anvil_e2_series.sv	73 total	47 code
rtl/anvil_e2_text_frontend_ple.sv	41 total	41 code
rtl/anvil_e2_top.sv	360 total	306 code
rtl/anvil_e2_vision_encoder_layer.sv	47 total	47 code

Production file	Physical lines	Code lines
rtl/anvil_e2_vision_patch_embed.sv	151 total	136 code
rtl/anvil_e2_vision_pool_proj.sv	190 total	148 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E002	design/OFFICIAL_WEIGHTS_EVALUATION.md
E006	rtl/anvil_e2_const_divide.sv
E014	rtl/anvil_e2_norm.sv
E020	rtl/anvil_e2_top.sv
E035	tests/official_weights.py
E070	.vulcan/engineering/run-0002/attempt-0014/stdout.log
E071	.vulcan/engineering/run-0002/attempt-0015/stdout.log
E072	.vulcan/engineering/run-0002/attempt-0016/stdout.log
E074	.vulcan/engineering/run-0002/rtl-timing-lint.json

Reference	Original project-relative record
E075	.vulcan/engineering/run-0002/quartus-sta.json
E076	.vulcan/engineering/run-0002/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).